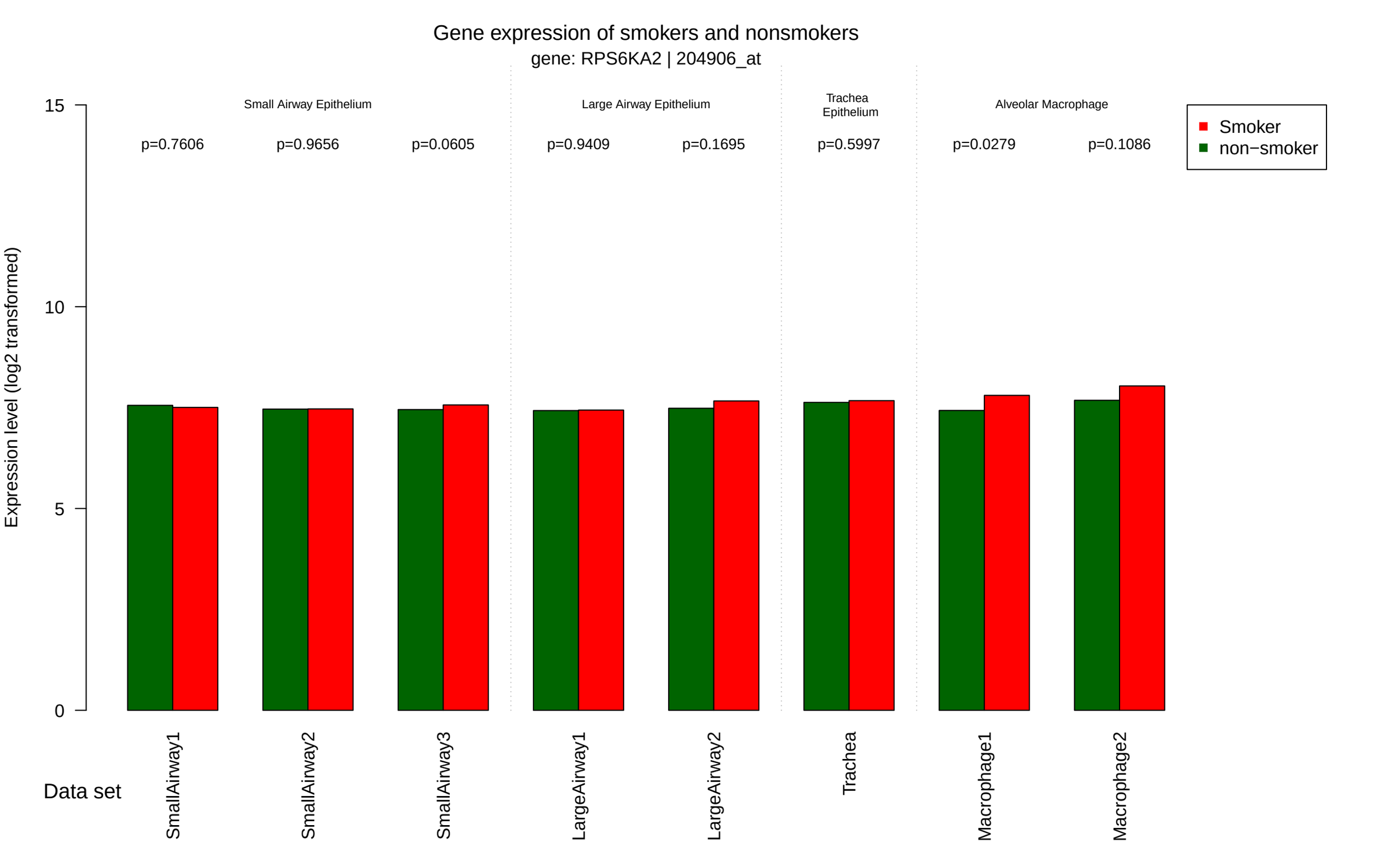
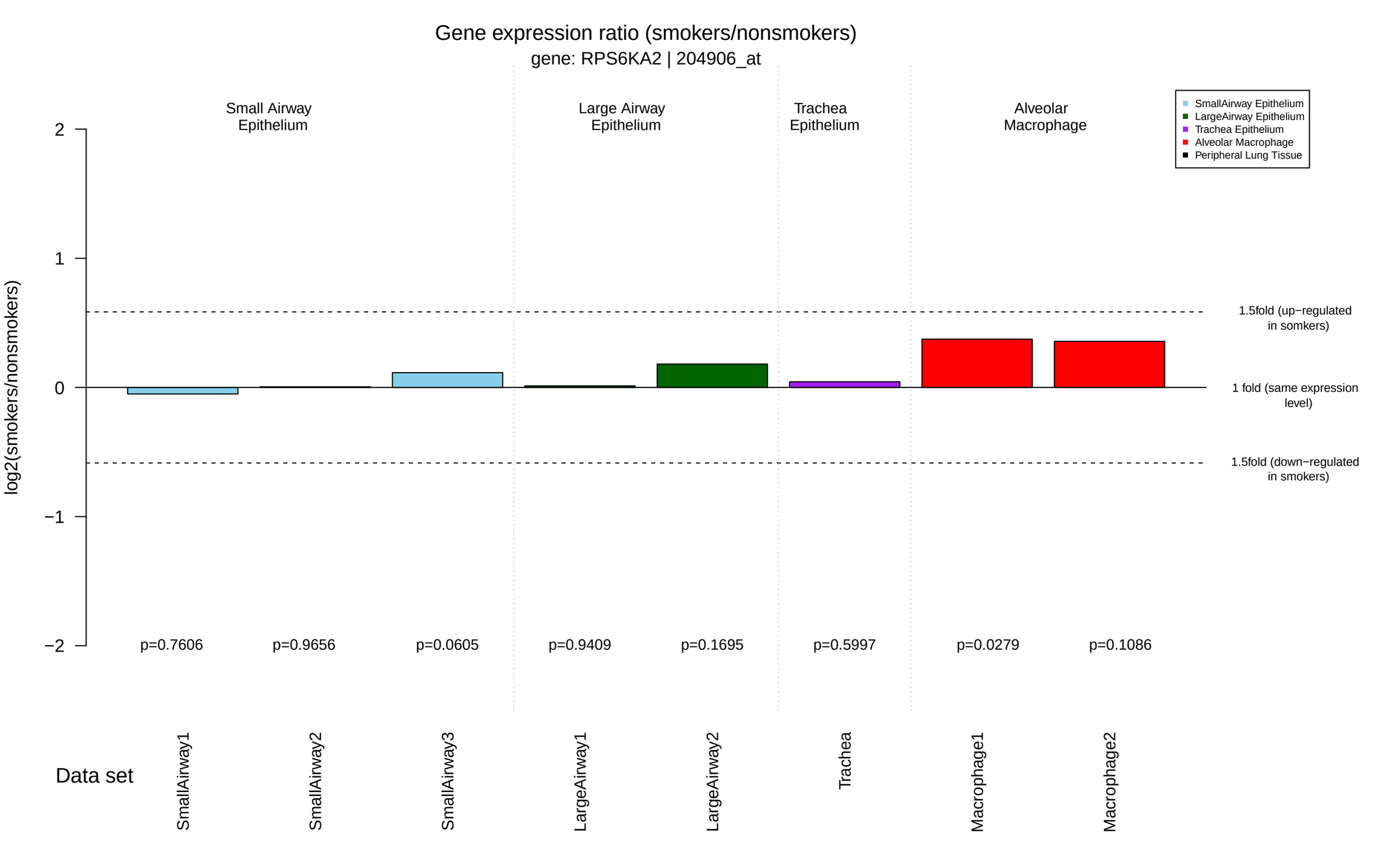
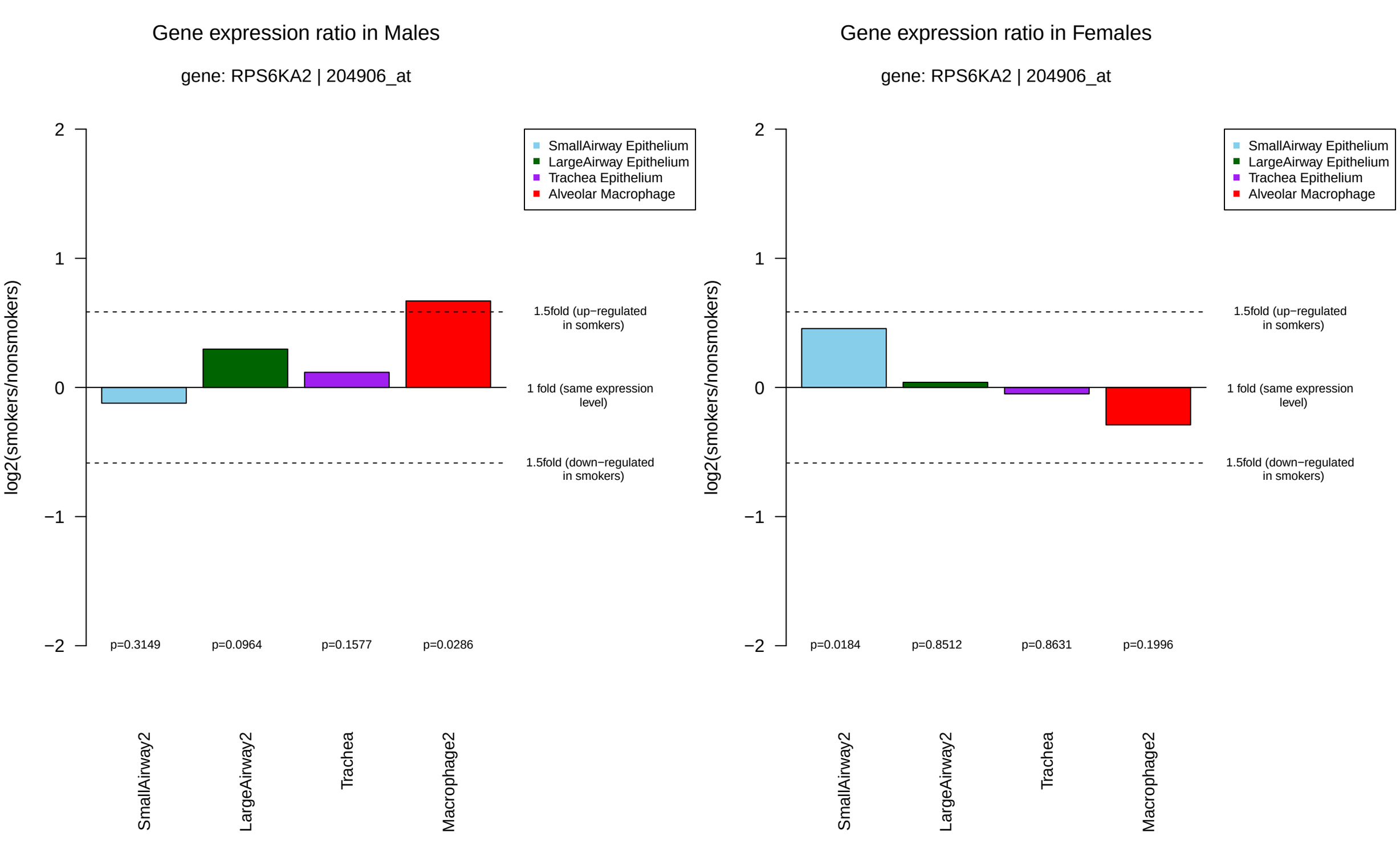
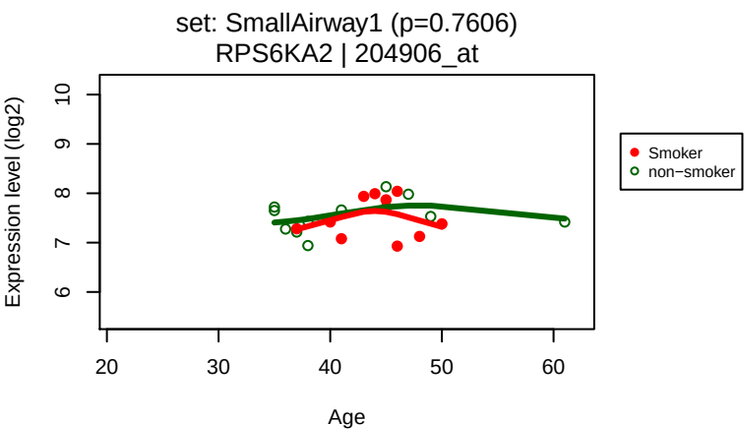
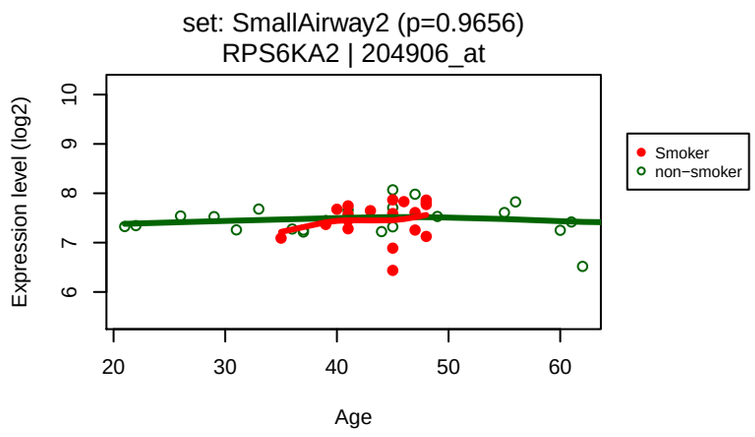
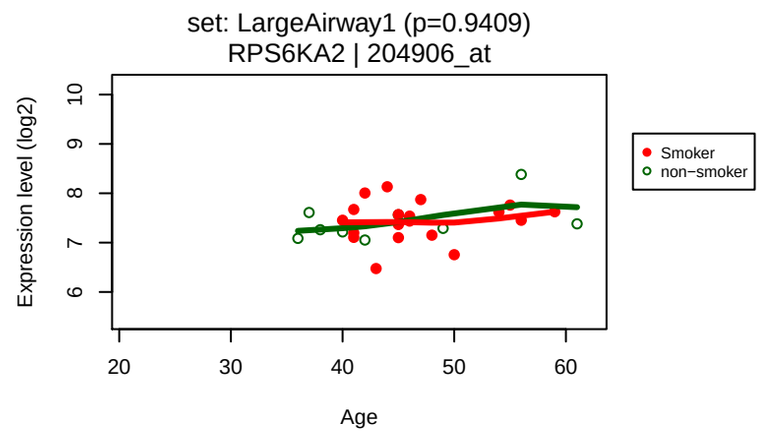

Gene expression of smokers and nonsmokers
gene: RPS6KA2 | 204906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7606
p=0.9656
p=0.0605
p=0.9409
p=0.1695
p=0.5997
p=0.0279
p=0.1086
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

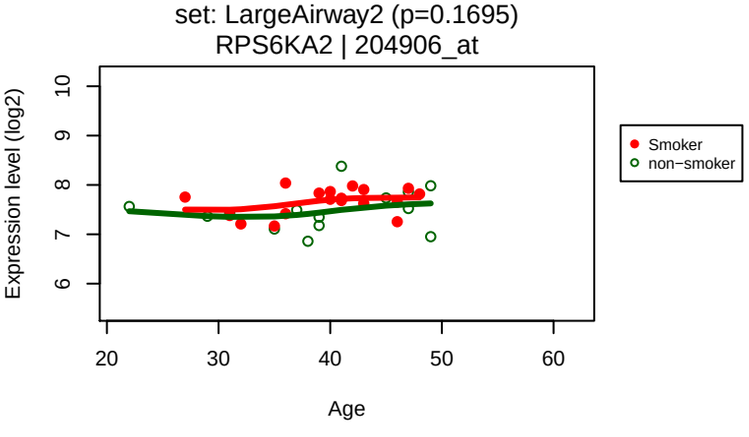

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA2 | 204906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7606
p=0.9656
p=0.0605
p=0.9409
p=0.1695
p=0.5997
p=0.0279
p=0.1086
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

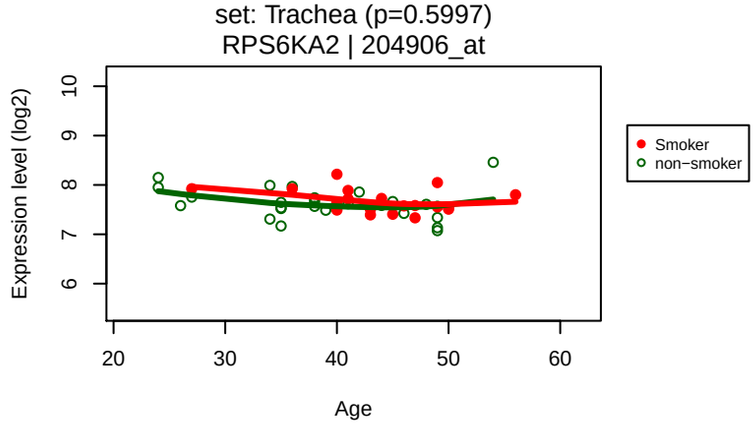

Gene expression ratio in Males
gene: RPS6KA2 | 204906_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3149
p=0.0964
p=0.1577
p=0.0286
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA2 | 204906_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0184
p=0.8512
p=0.8631
p=0.1996
SmallAirway2
LargeAirway2
Trachea
Macrophage2

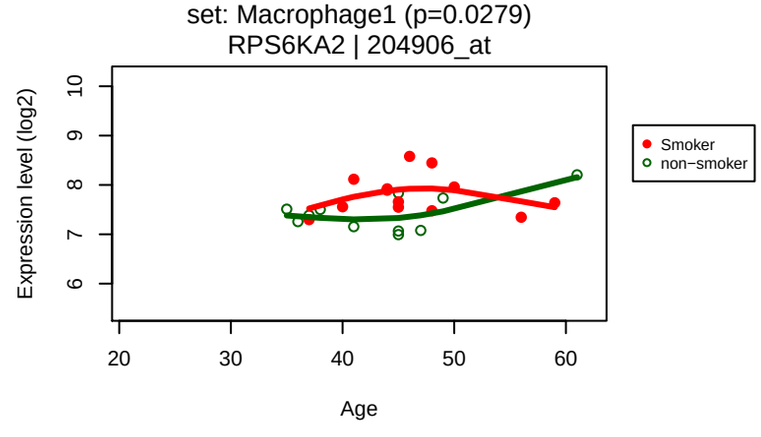

set: SmallAirway1 (p=0.7606)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker

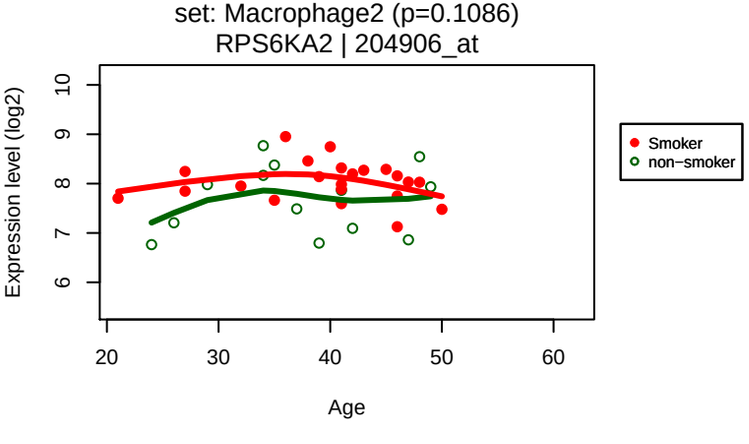

set: SmallAirway2 (p=0.9656)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker

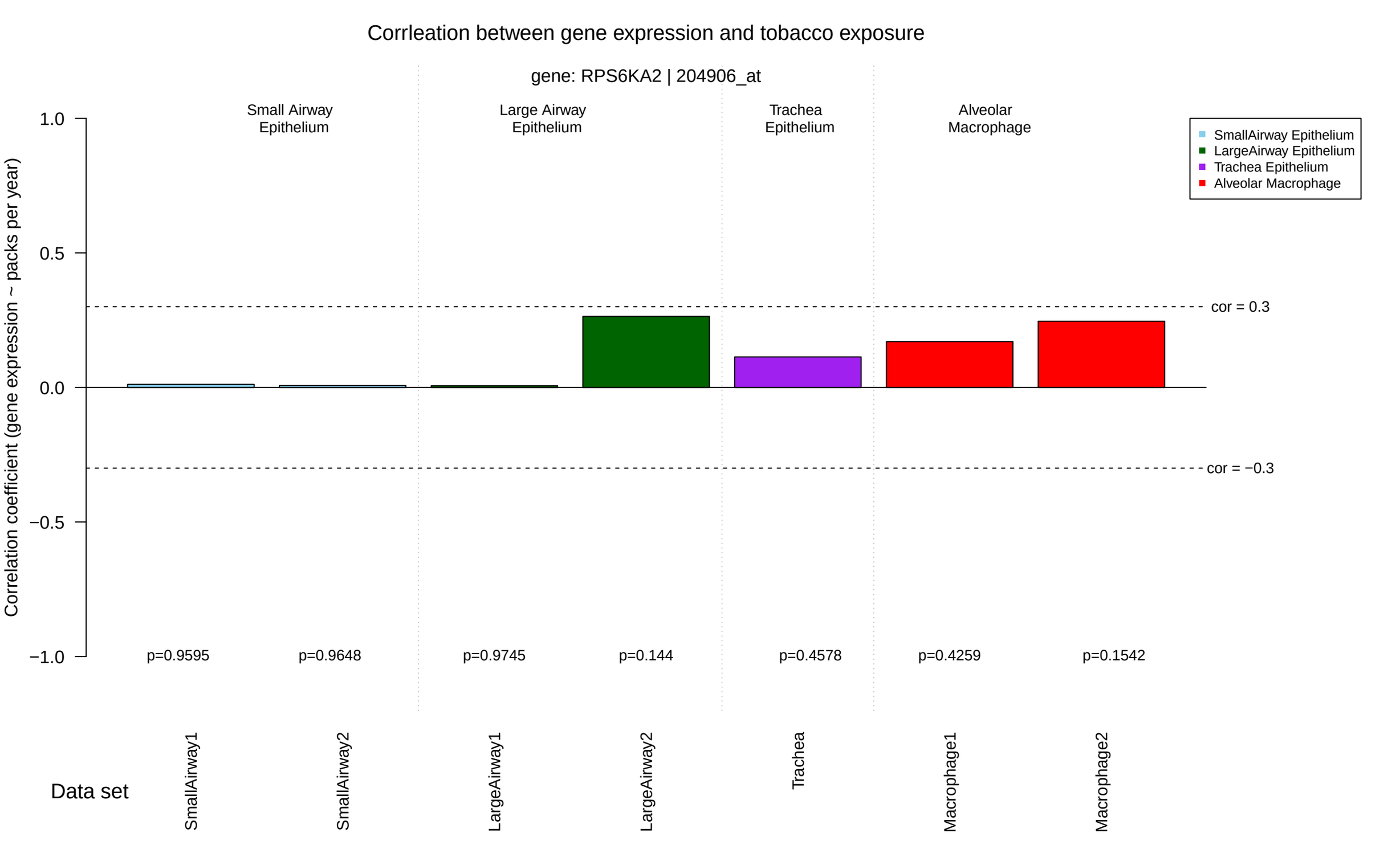

set: LargeAirway1 (p=0.9409)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1695)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5997)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0279)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1086)
RPS6KA2 | 204906_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA2 | 204906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9595
p=0.9648
p=0.9745
p=0.144
p=0.4578
p=0.4259
p=0.1542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2